Naming an unknown molecule from its mass spectra

How the current public model works, and what our first strategy session taught us · October 18, 2026

Kaggle "Enveda CASMI 2026 – Molecule ID From Mass Spectra", Sep 14 – Dec 14, 2026

1 The problem

A tandem mass spectrometer weighs a charged molecule (the **precursor ion**), shatters it by gas collisions and weighs the pieces. A **spectrum** lists the resulting peaks as (fragment m/z, intensity), m/z being mass over charge. For each of ~400 hidden natural-product-like molecules we get 1–16 spectra (median 3; Bruker timsTOF; 20/40/60 eV; ten possible adducts) and must return up to **25 ranked structures** (SMILES).

- **Score, MRR@25:** 1/rank of the first correct guess (rank 4 → 0.25; beyond 25 → 0), averaged. Correct = same atom connectivity (InChIKey first block; stereo ignored).
- **Noisy board:** the public leaderboard uses ~124 molecules (31%); one molecule going from miss to rank 1 moves it ~0.008 (standard error ~0.016).
- **Data and rules:** 2.54M training spectra of ~275,810 structures from 11 public libraries; offline notebook, ≤ 9 h; prizes need public, free data (no NIST) and reproducible weights.
- **Standing (Oct 17):** top 0.495; best public notebooks ~0.44–0.45; 3,486 teams.

Class	Meaning	Share*	Approach
1	has public reference spectra	16–28%	library search (MRR ~0.93)
2	in PubChem or COCONUT, no public spectra	20–45%	retrieve, rank isomers
3	novel: not in PubChem	39–52%	generate it

*Community leaderboard-probe estimates; the true split is hidden.

Jargon in plain words

Adduct – the charged form measured, e.g. $[M+H]^+$ = molecule + proton; it fixes M from the precursor m/z.
Fragment – a piece left after bonds break (one peak).
Isomers – different structures, same formula and mass.
Fingerprint – bits for substructures; *Tanimoto* = overlap.
Analog – a different molecule with a similar spectrum, e.g. the unknown minus one OH.
Forward model – predicts a structure's spectrum.
Ranker – orders candidates (LightGBM, LambdaRank).
ppm – parts-per-million mass error.

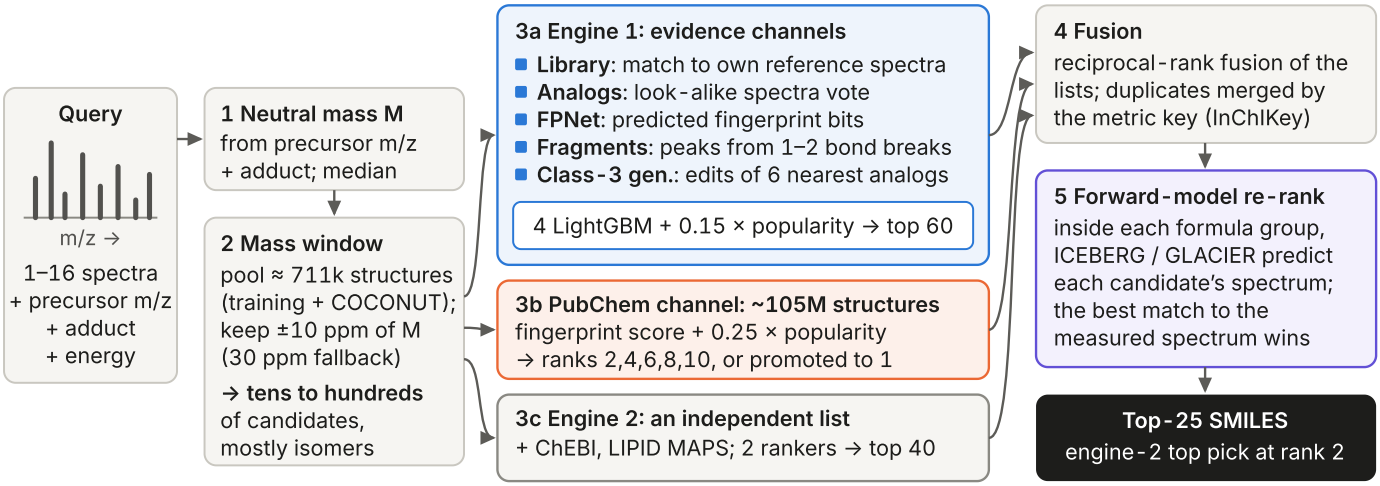


Figure 1. The current public model (community "E006" lineage; our fork scores 0.450). Mass cuts ~711k structures to a short list of mostly isomers; three engines score it with complementary evidence; fusion and forward models give the top 25.

2 How the current public model works

The best public notebook (we forked dmitriigluzdov's "From Spectra to Structures") stacks many community contributions into one **retrieve-then-rank** pipeline (Fig. 1):

1. **Shortlist by mass.** Candidates = pool structures within 10 ppm of M, mostly isomers, so the spectra must decide.
2. **Collect evidence (engine 1).** *Library match* (spectral-entropy similarity to the candidate's own reference spectra) solves class 1. *Analog propagation*: training spectra resembling the query (after a mass shift) vote for structurally similar candidates. *FPNet*, a transformer over peaks + metadata, predicts the fingerprint; candidates with agreeing bits score high. *Fragmentation*: how much peak intensity 1–2 bond breaks explain. Four LightGBM boosters combine ~100 features, plus a popularity bonus (log PubChem substance + literature counts).
3. **Look beyond the pool.** Fingerprint models screen ~105M PubChem structures; a proposal jumps to rank 1 if its fingerprint margin is large (> 150) or it is well documented, unless a strong library match (similarity ≥ 0.9) exists.
4. **Second opinion, fusion, check.** Engine 2 adds an independent list; fusion rewards candidates ranked high by either; forward models settle ties among isomers.

History: ~0.34 (mid-Sep) → 0.40 (engine fusion) → 0.44 (+ forward-model re-rank) → 0.45 (tuning).

Weak spots: almost no class-3 hits; many errors are the wrong same-formula isomer; most refinements were accepted on the noisy 124-molecule board without cross-validation; some bundled weights have uncertain licence or reproducibility (CC-BY-NC-SA; an unpublished ranker-row file), a prize risk.

3 Our strategy: what we tried, what we learned



Figure 2. Our five tracks (Oct 17–18). A–D ran in parallel and feed the decision in E; track B (orange) gave the main lesson. In A we kept the notebook's pinned Docker image because teams that ran copies on a newer image lost ~0.04.

B — Class 3, the largest untapped share

We built a stand-in for the unseen novel molecules: matching training structures against all 106M PubChem connectivity keys left **16,917 measured molecules that are not in PubChem**, our class-3 proxy. 45% of them split at one amide or ester bond into two PubChem-known pieces (N-acyl amino acids, bile-acid conjugates, fatty-acid esters). Our **NovelEngine** therefore emits only structures absent from PubChem, COCONUT and training, from two generators:

- **Analog derivatives:** find similar training spectra by *modified cosine* (which also matches peaks shifted by the mass difference Δ) and apply edits (hydroxylation, methylation, glycosylation, chain extension, …) whose mass change is Δ. The top-20 analogs reach the answer for 24% of the proxy.
- **Conjugates:** join a known acid and a known amine or alcohol (amide/ester/glycoside) whose masses add up; rank by diagnostic fragment ions of each half.

Our **site-localisation** score (Fig. 3) checks where an edit sits: derivative MRR on the proxy rose from 0.061 to 0.107. A learned LambdaRank ranker lifted proxy MRR from 0.124 to 0.269.

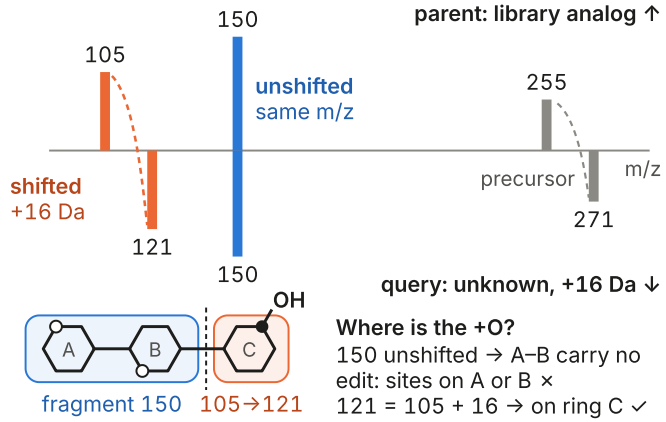


Figure 3. Site localisation (toy numbers). A library analog (up) mirrored against the query (down), 16 Da heavier (+O). Peak 150 stays, peak 105 moves to 121: fragment A–B is unmodified, so the +O sits on ring C (filled dot), not on A or B (open dots).

C — Bug hunt. A code review of the public pipeline found two **train/serve mismatches** (a feature computed one way in training, another at prediction time): (1) the "number of peaks" feature sees raw test peaks (median 230) but was trained on cleaned ones (median 52); (2) hidden-test positive-mode m/z carry an electron-mass correction (−0.000548 Da) missing from the enveda-180 training spectra. This shifts mass-error features by ~1.6 ppm at the median mass, versus a ~2.5 ppm typical gap between competing formulas. Two patches are pending.

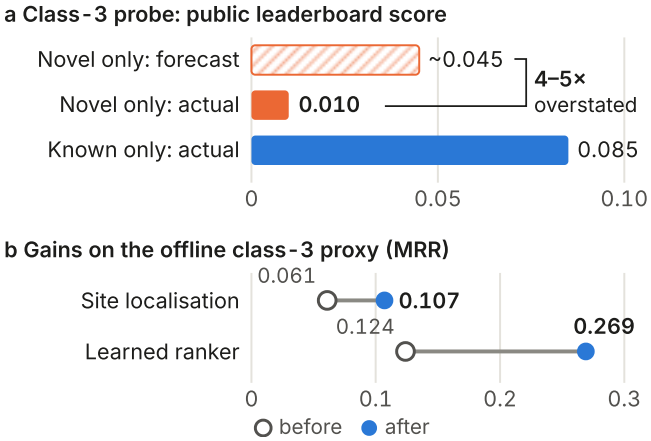


Figure 4. Proxy versus reality. (a) Novel-only candidates scored 0.010 vs a ~0.045 forecast; the engine's known-only list scored 0.085. (b) The proxy gains behind the forecast. Scales differ: (a) public test, (b) proxy only.

The lesson we put centre stage

A board probe with novel candidates only scored **0.010** against ~0.045 forecast: the proxy overstated class-3 reach ~4–5×. The engine works (known-only probe: 0.085), but class 3 is worth at most ~0.01 as an add-on, about one molecule at rank 1. **A proxy is a hypothesis about the test: validate it against the leaderboard early.** As the board is noisy (±0.016), we also need offline validation that mimics the test (track D).

D — Compliance-first own pipeline. Prize entries need models reproducible from permitted data, so we build our own: a 726,198-structure pool (training + COCONUT), library, analog, fragment and COCONUT-documentation channels, and a **dual encoder**. One MLP embeds the spectrum (1-Da peak bins, neutral losses, adduct/energy/instrument), another the molecule (2,215 fingerprint bits); a contrastive loss pulls the true structure closer than up to 64 same-mass "hard negatives". After one CPU epoch: **MRR 0.51** (top-1 36%) among same-mass candidates on held-out structures. A LambdaRank ranker trains on simulated queries from 8,000 structures (class 1: hide the query's spectra; class 2: hide all its spectra) and is validated on 1,500 disjoint ones, using the *measured* mass (the true one would leak the answer) and the electron correction.

E — Next. Compare variants on the board, finish the own pipeline (target ≥ 0.45, fully compliant), add class-3 candidates only where the main list is weak, and pick the two finals by validation + board. If the rules are enforced, strong compliant entries may outrank non-reproducible ones.